BEING A LOCAL COMPLETE INTERSECTION IS WELL DEFINED FOR QUOTIENTS OF REGULAR LOCAL RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Regular local rings are Cohen-Macaulay. Invoked at the start of the counted proof (line 36267). Allows the number of generators to be compared with the codimension via the theory of Cohen-Macaulay modules.
- The quotient of a regular local ring by a regular element. Used in the induction at the end of the counted proof (lines 36289-36292). Reduces both the dimension and the minimal number of generators by one.

Source reference. Original lemma 00NR: sources/stacks-algebra/pinned-original-source.tex, source lines 25967–25967. Navigation pointer for \ref{lemma-regular-quotient-regular} used inside the counted span (line 36270); the argument that consumes it is retained in the span.

Source reference. Original lemma 00NQ: sources/stacks-algebra/pinned-original-source.tex, source lines 25949–25949. Navigation pointer for \ref{lemma-regular-ring-CM} used inside the counted span (line 36267); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00N6: sources/stacks-algebra/pinned-original-source.tex, source lines 25269–25269. Navigation pointer for \ref{proposition-CM-module} used inside the counted span (line 36269); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $A \to B \to C$ be surjective local ring homomorphisms. Assume A and B are regular local rings. The following are equivalent*

(1) $\mathrm{Ker}(A \to C)$ *is generated by a regular sequence,*
(2) $\mathrm{Ker}(A \to C)$ *is generated by* $\dim(A) - \dim(C)$ *elements,*
(3) $\mathrm{Ker}(B \to C)$ *is generated by a regular sequence, and*
(4) $\mathrm{Ker}(B \to C)$ *is generated by* $\dim(B) - \dim(C)$ *elements.*

Proof. A regular local ring is Cohen-Macaulay, see Lemma 00NQ. Hence the equivalences (1) $\Leftrightarrow$ (2) and (3) $\Leftrightarrow$ (4), see Proposition 00N6. By Lemma 00NR the ideal $\mathrm{Ker}(A \to B)$ can be generated by $\dim(A) - \dim(B)$ elements. Hence we see that (4) implies (2).

It remains to show that (1) implies (4). We do this by induction on $\dim(A) - \dim(B)$. The case $\dim(A) - \dim(B) = 0$ is trivial. Assume $\dim(A) > \dim(B)$. Write $I = \mathrm{Ker}(A \to C)$ and $J = \mathrm{Ker}(A \to B)$. Note that $J \subset I$. Our assumption is that the minimal number of generators of I is $\dim(A) - \dim(C)$. Let $\mathfrak{m} \subset A$ be the maximal ideal. Consider the maps

$$J/\mathfrak{m}J \to I/\mathfrak{m}I \to \mathfrak{m}/\mathfrak{m}^2$$

By Lemma 00NR and its proof the composition is injective. Take any element $x \in J$ which is not zero in $J/\mathfrak{m}J$. By the above and Nakayama's lemma x is an element of a minimal set of generators of I. Hence we may replace A by A/xA and I by I/xA which decreases both $\dim(A)$ and the minimal number of generators of I by 1. Thus we win. $\square$